

An-Chi Liu

SOFTWARE ENGINEER · SYSTEM SOFTWARE & ROBOTIC & AUTONOMOUS DRIVING PROFESSION

Taiwanese, based in Tokyo, Japan (Permanent Resident)

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Work Experience

T2, Inc.

Senior Software Engineer

- Led a **two-person project** to develop a **QA cluster platform** from scratch for running T2's simulation tools. Designed the system architecture and implemented the software stack with **Kubernetes** and **Docker**, the backend in **Golang**, and the frontend in **React**, all built with **AI-driven development** practices. Also designed and built the underlying cluster hardware.
- Designed and implemented an **OpenDrive-like** mapping system and an **OpenScenario-like** scenario definition system, supporting various types of maps and scenarios to perform closed-loop simulation with the autonomous driving planning system.

Hybrid
Nov. 2025 – Present

Datra AI

Technical Co-founder

- Co-founded an AI startup building an **AI chatbot** with an inbox system for **Shopify** merchants in the skincare industry.
- Led end-to-end product development, including infrastructure, system architecture, backend (**Golang**), frontend (**React**), **AI engineering**, and deployment on **AWS** with **Docker**.
- Built an **AI system** integrating various **LLMs** (e.g., OpenAI, Claude), with optimized **prompt engineering** and retrieval-augmented generation (**RAG**) techniques.
- Collaborated with the business co-founder and managed up to **2 part-time developers** who assisted with product development and maintenance.

Remote
Feb. 2025 – Sep. 2025

Mujin, Inc.

Backend Software Engineer

- Developed a universal **network device discovery service** in **C++** using **PcapPlusPlus** that detects IP devices (robots, industrial cameras, etc.) on the network within seconds.
- Developed a **real-time live streaming and playback system** supporting over **100** surveillance cameras with latency under **50 ms**, based on **FFmpeg**, **WebRTC**, and **Golang**.
- Maintained the robot and warehouse systems written in **C++** and **Python**, using **ZMQ** and **Qt/QML**.
- Implemented new **web APIs (Golang)** and **ZMQ APIs (C++)** to support frontend projects and internal system services.
- Designed internal web-based UI tools using **Figma** and **React**.

Tokyo, Japan
Sep. 2022 – May 2025

Education

National Yang Ming Chiao Tung University (NYCU)

M.S. in Computer Science and Engineering

- GPA: 4.11/4.3. Thesis: **Offworker: An Offloading Framework for Parallel Web Applications**, published in **WISE 2022** (top-tier conference).
- Received the **Best Master's Thesis Award** from the Institute of Information & Computing Machinery, Taiwan (2022; one of 13 awardees).

Hsinchu, Taiwan
Sep. 2019 – May. 2022

National Taiwan University (NTU)

B.S. in Bio-Industrial Mechatronics Engineering

Taipei, Taiwan
Sep. 2014 – Jun. 2019

Open-Source Projects

PcapPlusPlus

- PcapPlusPlus is a multi-platform **C++ library** for capturing, parsing, and crafting **network packets**, with **3k stars** on GitHub.
- **One of five core maintainers** of PcapPlusPlus since 2024.
- Major contributions include fixing performance issues, establishing a more efficient CI/CD pipeline, adding new features, and improving documentation.

modmesh

- modmesh is a high-performance toolkit for solving partial differential equations, built with **C++** and **pybind11** bindings, with **61 stars** on GitHub.
- Major contributor, with contributions including a **low-level NumPy-like array**, **polygon boolean operations** and **data structures**, and **drawing APIs** for the pilot GUI tool.
- Gave a talk at **PyCon JP 2024**: "Crafting Your Own NumPy: Do More in C++ and Make It Python".

Skills

Programming

Proficient: C++11, C++14, C++17, **Experience:** Golang, Python 3, JavaScript, TypeScript, Node.js

Frameworks/Tools

Unix/Linux, CMake, ZMQ, CI/CD, Kubernetes, Docker, AWS, GDB, Git, pybind11, Qt/QML, REST API, GraphQL, FFmpeg, GoogleTest, WebRTC, React, etc.

AI Skills

Claude Code, Codex, LLM integration (OpenAI, Claude, etc.), Prompt engineering, RAG, MCP, etc.

Languages

Mandarin Chinese (Native), English (Business), Japanese (Limited Conversational)